Identity Security Guide — Threat Landscape & Defense

> 🎯 **Goal**: Understand why identity is the #1 attack vector, how modern adversaries exploit identity weaknesses, and how to defend AI agents and enterprise systems using CrowdStrike-led identity security practices.

> ⏱️ **Time**: 1.5–2h reading

> 📦 **Deliverable**: A clear understanding of identity-based attack patterns, CrowdStrike's Falcon Identity Security architecture, and a practical defense playbook for securing both human and agent identities.

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Why Identity Is the #1 Attack Vector

The CrowdStrike 2025 Global Threat Report reveals a fundamental shift: **adversaries no longer need malware — they log in**.

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│              THE IDENTITY CRISIS (CrowdStrike 2025 GTR)          │
│                                                                  │
│   79%  of attacks are MALWARE-FREE                               │
│         (up from 40% in 2019 — attackers use stolen credentials) │
│                                                                  │
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│   442%  increase in VISHING attacks                                    │
│         (voice phishing, H1→H2 2024)                                   │
│                                                                        │
│   50%   increase in ACCESS BROKER activity                             │
│         (criminals selling valid credentials on dark markets)          │
│                                                                        │
│   68%   of identity attacks BYPASS MFA                                 │
│         (session hijacking, MFA fatigue, SIM swap)                     │
│                                                                        │
│   48min average BREAKOUT TIME                                          │
│         (from initial access to lateral movement)                      │
│         Fastest observed: 51 SECONDS                                   │
│                                                                        │
│   71%   of interactive intrusions involved                             │
│         VALID ACCOUNT abuse                                            │
│                                                                        │
│   136%  increase in CLOUD INTRUSIONS (H1 2025)                         │
│         (misconfigurations + stolen credentials)                       │
│                                                                        │
│  Bottom line: If you control the identity, you control everything.     │
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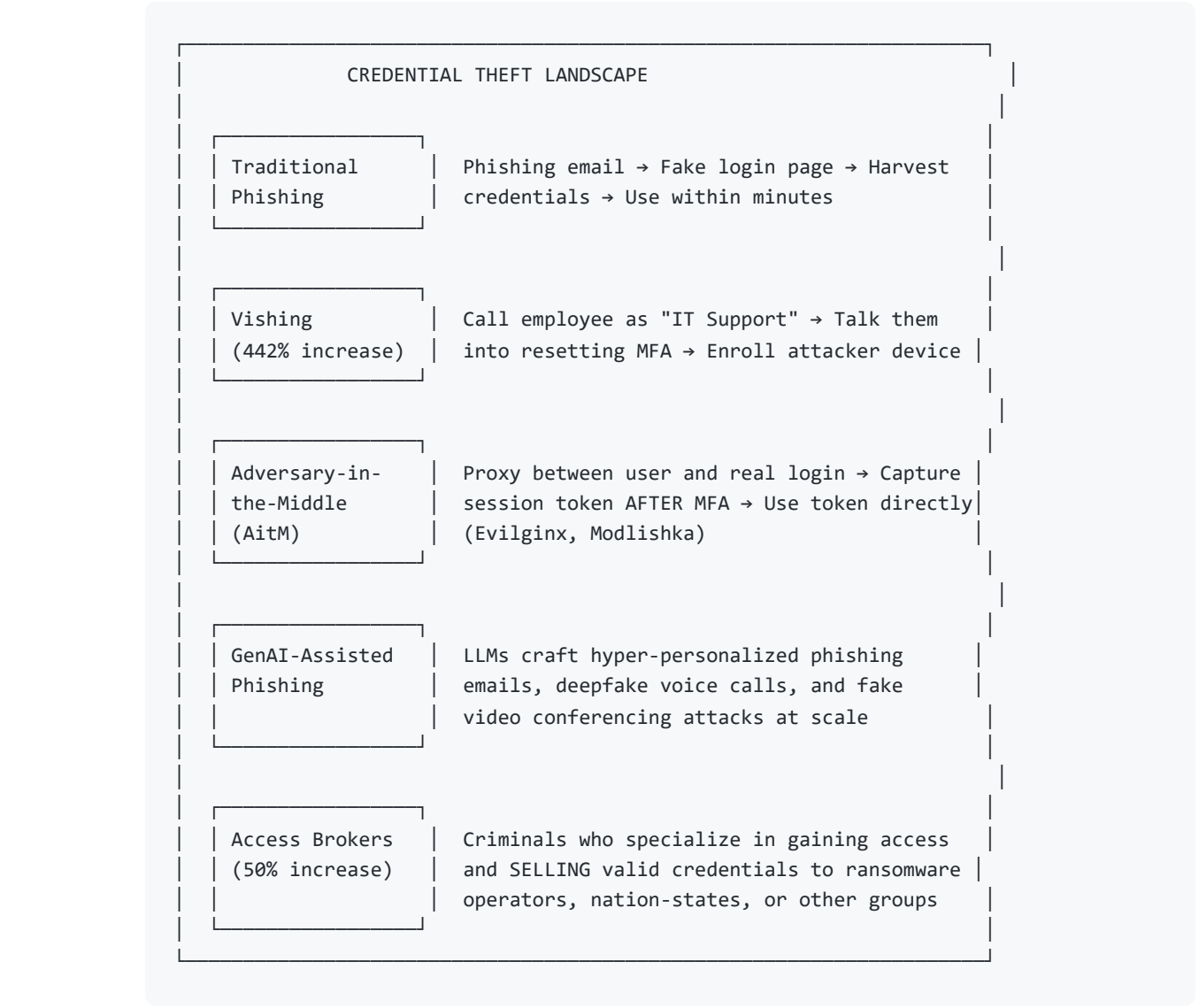
Why Attackers Prefer Identity Over Malware

Factor	Malware-Based Attack	Identity-Based Attack
Detection	EDR catches known signatures/behaviors	Legitimate credentials look normal
Speed	Must install, execute, evade	Just log in — instant access
Persistence	Malware can be quarantined	Compromised account persists until revoked
Lateral movement	Requires exploits for each hop	Valid credentials move freely
Cost to attacker	Must develop or purchase malware	Buy credentials for ~$10 on dark web
Forensics	Leaves artifacts (files, processes)	Looks like normal user activity

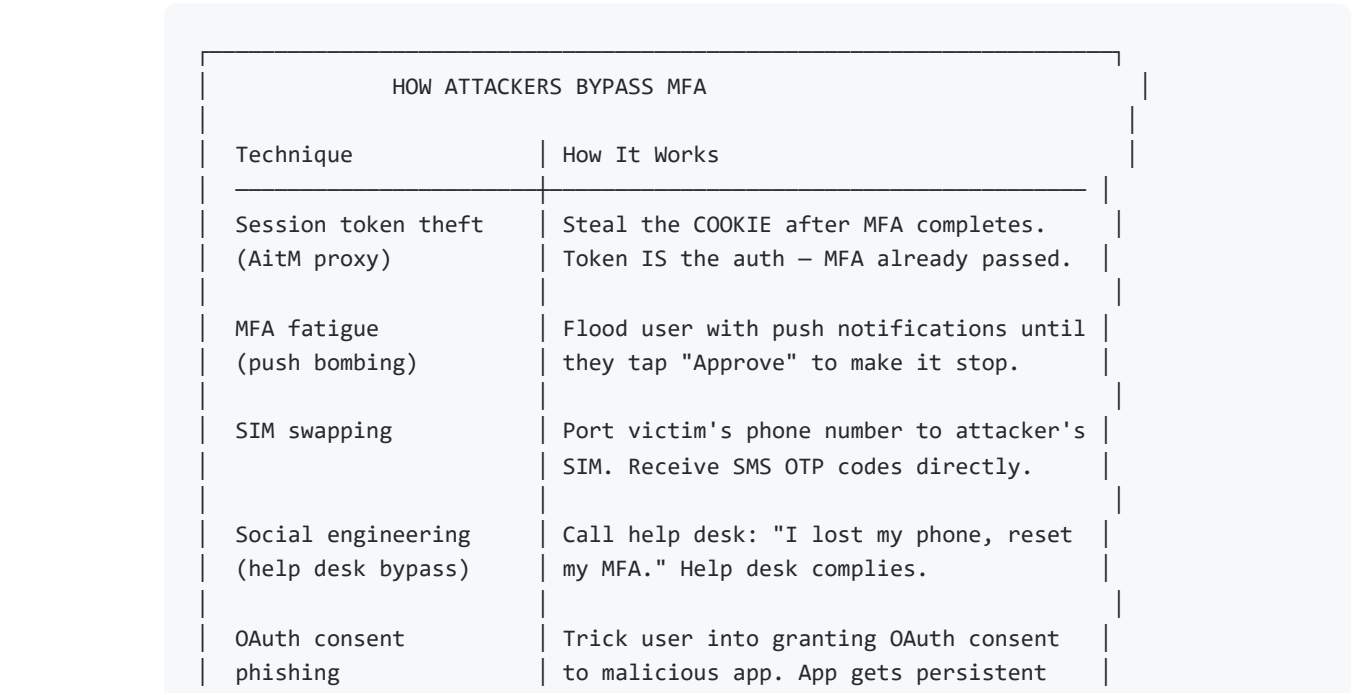
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Identity Attack Techniques

1. Credential Theft & Phishing



2. MFA Bypass Techniques (68% Success Rate)

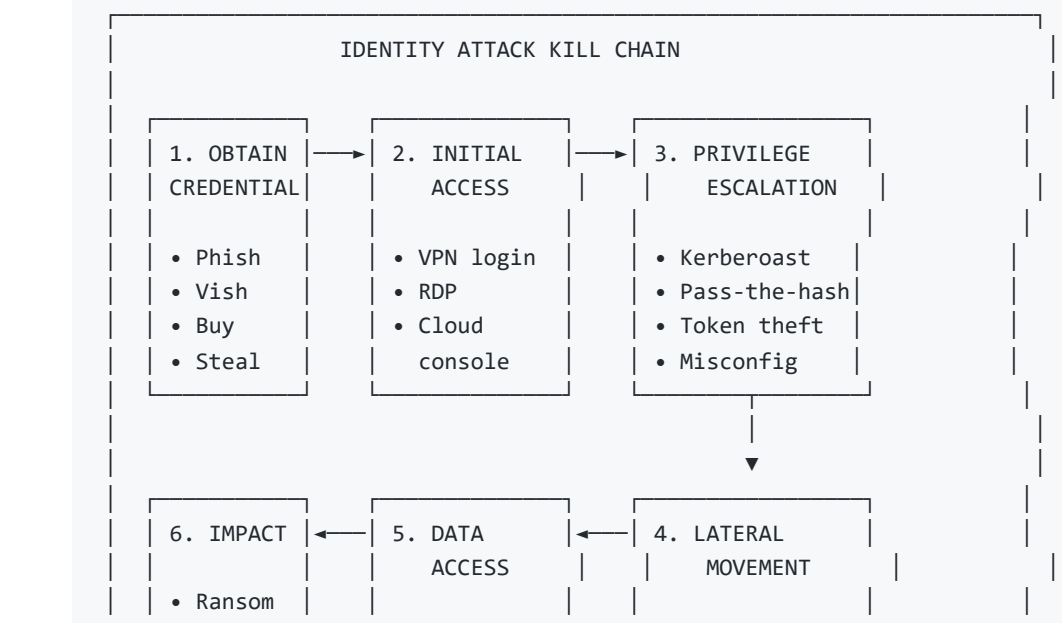


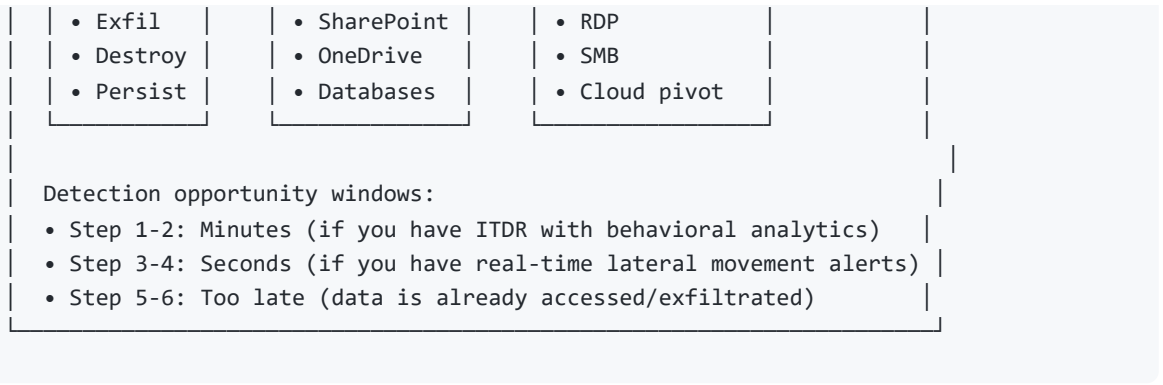
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│                          │ access token — no MFA needed for API.    │
│                          │                                          │
│  Recovery code theft     │ Obtain backup codes through phishing     │
│                          │ or data breach. Bypass all MFA.          │
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3. Lateral Movement (48min Average Breakout)

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│              IDENTITY-BASED LATERAL MOVEMENT                     │
│                                                                  │
│  Timeline of a real attack:                                      │
│                                                                  │
│  T+0:00   Attacker logs in with stolen credentials (looks normal)  │
│  T+0:02   Enumerates Active Directory (BloodHound, SharpHound)     │
│  T+0:05   Discovers service account with Domain Admin membership   │
│  T+0:08   Kerberoasts the service account (offline cracking)       │
│  T+0:15   Cracks weak service account password                     │
│  T+0:18   Uses Domain Admin to access file servers                 │
│  T+0:25   Discovers cloud admin credentials in a config file       │
│  T+0:30   Pivots to Azure/M365 using cloud admin account           │
│  T+0:35   Exfiltrates SharePoint data                              │
│  T+0:45   Deploys ransomware across domain                         │
│  T+0:48   ← Average breakout time                                  │
│                                                                  │
│  Key: Every step used VALID CREDENTIALS. No malware involved       │
│  until ransomware deployment at the very end.                    │
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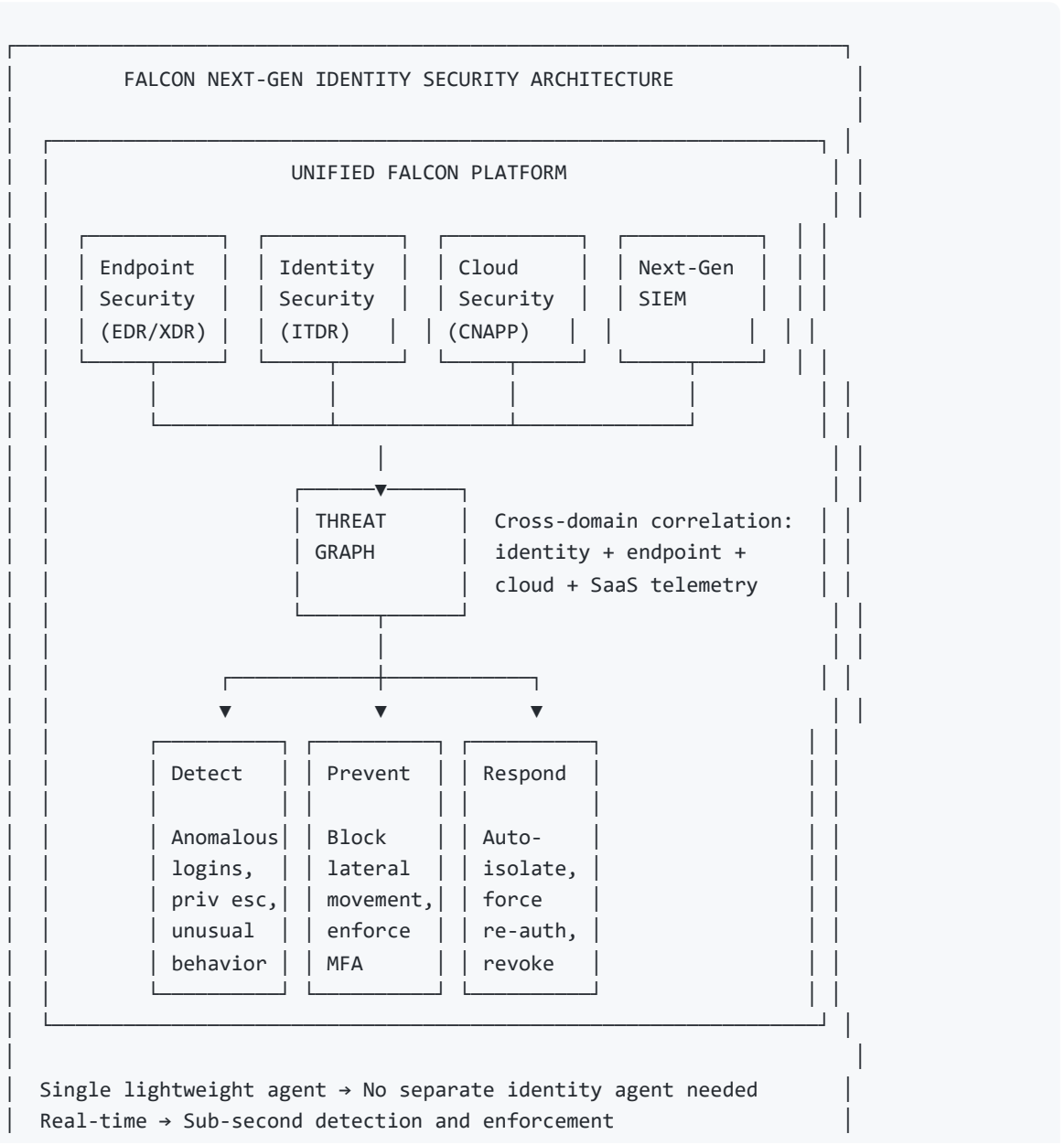
The Identity Attack Kill Chain





CrowdStrike Falcon Next-Gen Identity Security

CrowdStrike's Falcon platform provides unified identity security across the entire attack chain. Here's the architecture:

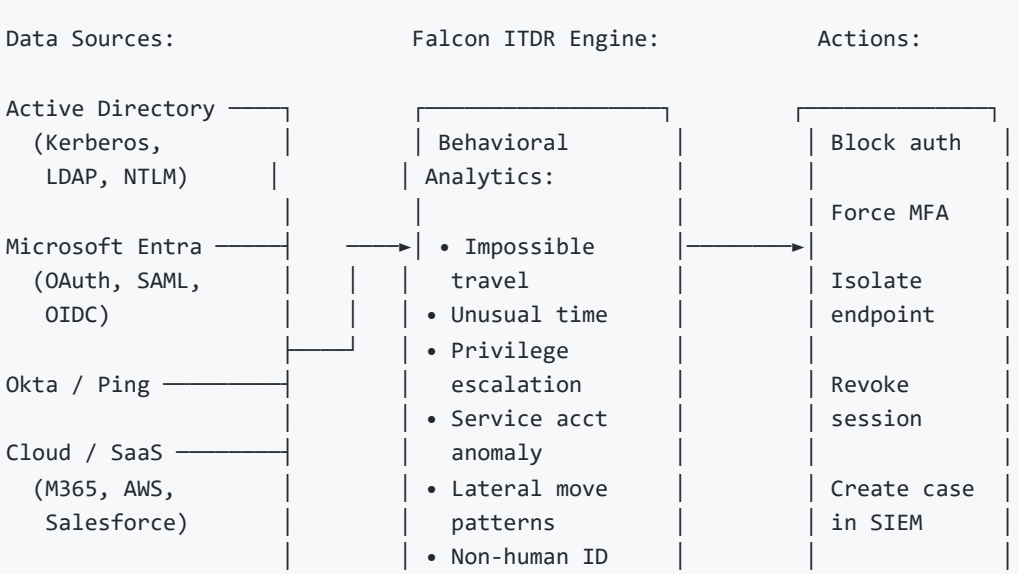


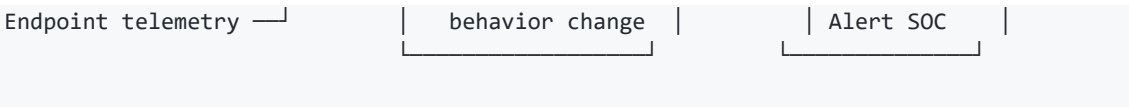
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│  Agentic AI → Automated triage, investigation, and response        │
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Key Components

Component	What It Does	Why It Matters
Identity Threat Detection & Response (ITDR)	Real-time monitoring of authentication events, credential usage, and privilege changes across AD, Entra, Okta, and SaaS apps	Detects credential misuse before lateral movement succeeds
FalconID	Phishing-resistant, passwordless MFA using FIDO2 + endpoint/identity risk signals	Eliminates the #1 attack vector (stolen passwords + MFA bypass)
Privileged Access Management	Zero-standing-privileges: JIT access grants, auto-revocation, risk-based approval	Eliminates permanently over-privileged accounts
Identity-Driven Case Management	Auto-correlates identity + endpoint + cloud detections into unified investigation cases	Full attack chain visibility in one place
Agentic AI for Enforcement	Autonomous AI agents that triage alerts, investigate context, and take enforcement actions	Reduces human SOC analyst burden, sub-second response

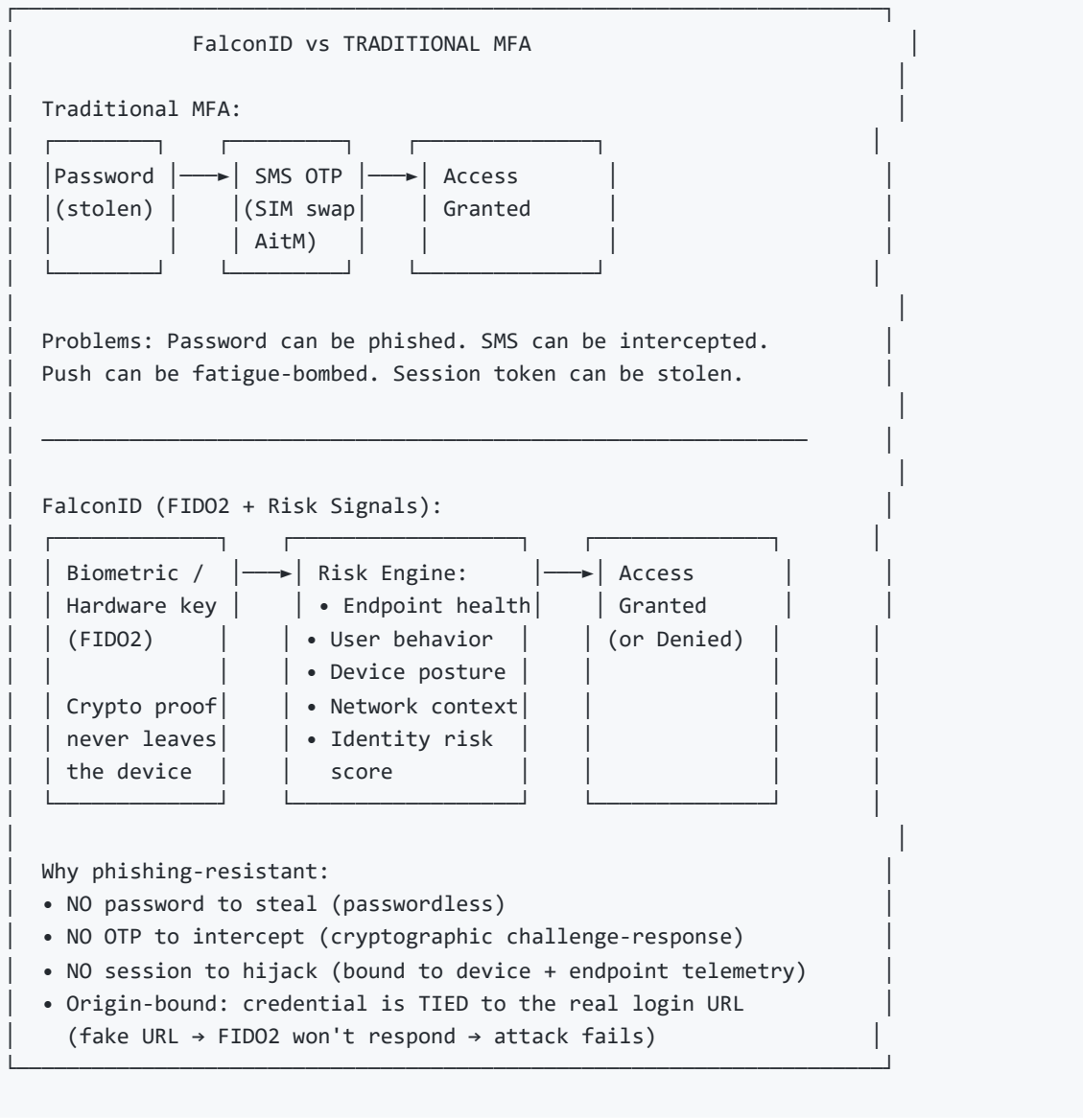
How ITDR Works (Detection Pipeline)





FalconID: Phishing-Resistant Passwordless MFA

FalconID is CrowdStrike's answer to the MFA bypass problem (68% of attacks bypass traditional MFA).



Protecting AI Agent Identities

AI agents are what CrowdStrike calls **"superhuman identities"** — they operate 24/7, often with elevated privileges, and are increasingly targeted.



Defending Agent Identities

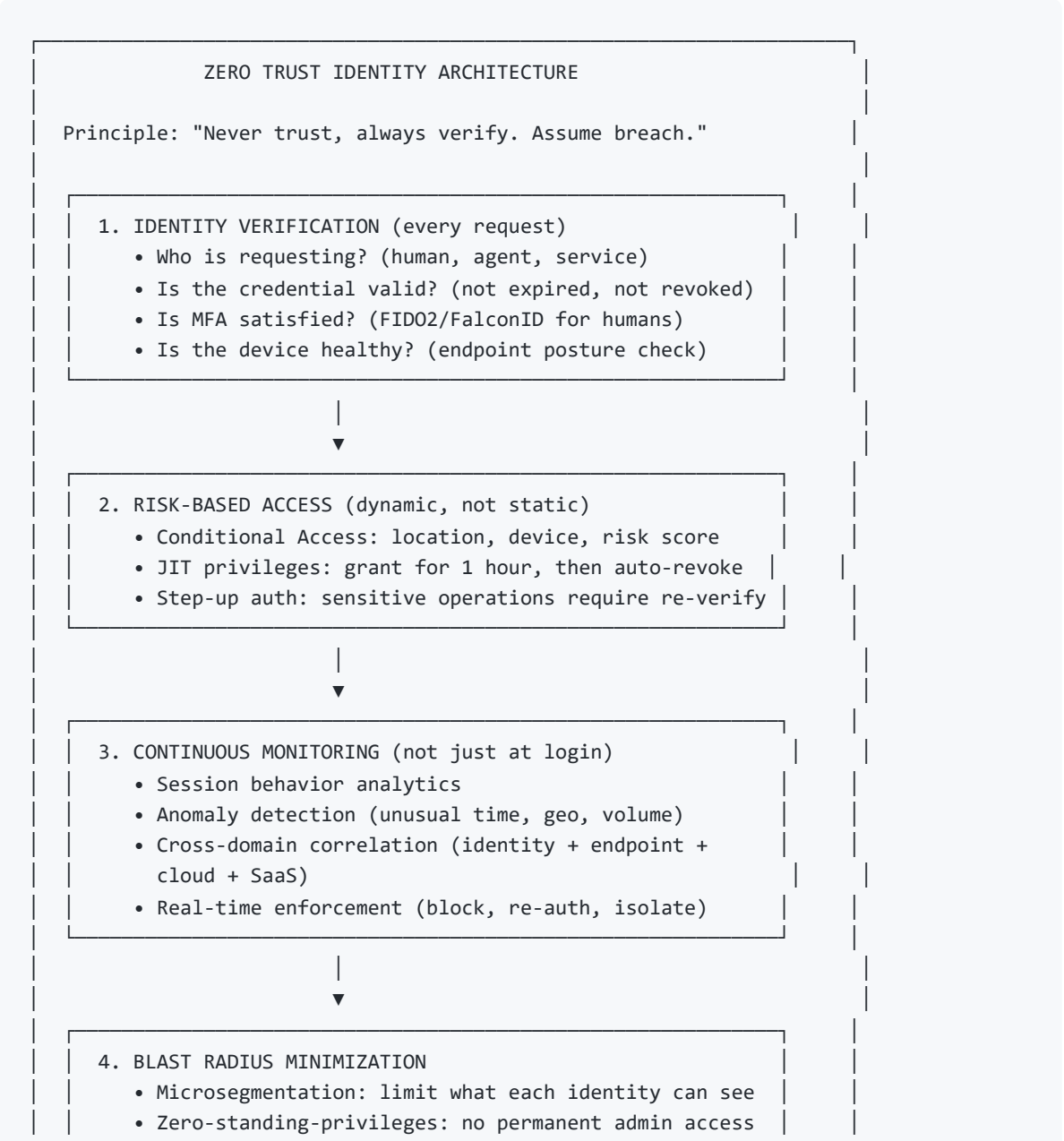
Defense	How To Implement	CrowdStrike Role
Use secretless auth	Managed Identity, Workload ID Federation, SPIFFE (see Authentication Guide)	N/A — infrastructure-level
Monitor non-human behavior	Baseline agent activity patterns; alert on anomalies (new APIs, new IPs, unusual volumes)	Falcon ITDR detects non-human identity anomalies
Least privilege	Each agent gets only the permissions it needs. Review quarterly	Falcon PAM enforces zero-standing-privileges
Lifecycle management	Inventory all agents. Decommission unused ones. Track owner and purpose	Entra Agent ID + Falcon catalog integration

Conditional Access	Block agent access from unexpected networks, times, or geolocations	Entra Conditional Access + Falcon endpoint posture
Audit everything	Log every agent action with traceable identity	Falcon SIEM + Entra audit logs
Rotate credentials	If secrets are used, rotate frequently (<90 days). Prefer FIC	Falcon alerts on long-lived credentials

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Identity Security Architecture

A modern identity security architecture follows **zero trust** principles:



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│  │     • Auto-containment: isolate compromised identities     │     │
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Defense Playbook

Practical Controls by Priority

Priority	Control	Implementation	Addresses
P0	Deploy phishing-resistant MFA (FIDO2)	FalconID, Windows Hello, YubiKey	68% MFA bypass, all phishing
P0	Eliminate standing privileges	JIT access via PAM, auto-revoke after use	Privilege escalation, lateral movement
P1	Monitor identity telemetry	Falcon ITDR across AD, Entra, Okta, SaaS	All identity-based attacks
P1	Protect service accounts	Rotate passwords, monitor usage, restrict login sources	Kerberoasting, lateral movement
P1	Inventory non-human identities	Catalog all agents, service accounts, API keys. Track owners	Agent sprawl, orphaned credentials
P2	Enforce Conditional Access	Block legacy auth, require device compliance, geo-restrict	Account takeover from unexpected locations
P2	Detect lateral movement	Cross-domain correlation (endpoint + identity + network)	48min breakout window attacks
P2	Secure help desk processes	Require identity verification before MFA resets	Vishing (442% increase)
P3	Train users on vishing	Regular simulated phishing + vishing campaigns	Social engineering
P3	Monitor access broker markets	Threat intelligence feeds for leaked credentials	Credential purchasing

Detection Strategies

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│              HIGH-SIGNAL IDENTITY DETECTIONS                         │
│                                                                      │
│  Detection                      | Signal                             │
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│  Impossible travel               | Login from NYC, then London 10     │
│                                  | minutes later                      │
│                                  |                                    │
│  Dormant account activation      | Service account unused for 90      │
│                                  | days suddenly makes API calls      │
│                                  |                                    │
│  Privilege escalation            | Non-admin account adds self to     │
│                                  | Domain Admins group                │
│                                  |                                    │
│  Unusual authentication method   | Account that always uses SSO       │
│                                  | suddenly uses password + OTP       │
│                                  |                                    │
│  Token anomaly                   | OAuth token used from different    │
│                                  | IP than where it was issued        │
│                                  |                                    │
│  Mass enumeration                | One account queries 500+ AD        │
│                                  | objects in 60 seconds              │
│                                  |                                    │
│  Agent behavior change           | AI agent suddenly accesses         │
│                                  | resources outside its normal       │
│                                  | pattern (new APIs, new tables)     │
│                                  |                                    │
│  Cross-domain pivot              | On-prem credentials used to        │
│                                  | access cloud resources for         │
│                                  | first time (hybrid attack)         │
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Interview Angle

Key Points to Articulate

1. "**Identity is the new perimeter**" — 79% malware-free means endpoint-only security is insufficient. You must protect the identity layer
2. "**MFA is necessary but not sufficient**" — 68% bypass rate. Need phishing-resistant MFA (FIDO2), not just push/SMS. Also need continuous session monitoring
3. "**48-minute breakout time**" — automated detection and response is essential. Human SOC analysts can't react fast enough alone
4. "**Cross-domain correlation is critical**" — identity attacks span AD → cloud → SaaS. Single-domain visibility misses the attack chain. CrowdStrike unifies endpoint + identity + cloud telemetry

5. **"AI agent identities are the next frontier"** — superhuman identities with persistent high privileges. Need the same governance as human accounts: least privilege, monitoring, lifecycle management, conditional access

Common Interview Questions

Question	Key Talking Points
"How do identity-based attacks work?"	Credential theft (phishing, vishing, access brokers) → initial access → privilege escalation → lateral movement. 79% malware-free per CrowdStrike 2025 GTR
"Why does MFA fail?"	AitM proxies steal session tokens, MFA fatigue, SIM swap, help desk social engineering. Solution: FIDO2 (origin-bound, phishing-resistant)
"What is ITDR?"	Identity Threat Detection & Response. Real-time monitoring of auth events across AD/Entra/Okta/SaaS. Behavioral analytics detect anomalies. Auto-enforcement (block, re-auth, isolate)
"How do you secure AI agents?"	Secretless auth (MI/Federation), least privilege, behavioral monitoring, lifecycle management, conditional access, audit trail
"Explain zero trust for identity"	Never trust, always verify. Every request verified (identity + device + context). JIT privileges. Continuous monitoring. Blast radius minimization

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References & Links

CrowdStrike Reports & Research

- CrowdStrike 2025 Global Threat Report (PDF) — Full annual threat landscape report
- How to Navigate the 2025 Identity Threat Landscape — CrowdStrike blog post with identity-specific analysis
- Falcon Next-Gen Identity Security Press Release — Official announcement of ITDR, FalconID, PAM innovations
- NAND Research: CrowdStrike Falcon Next-Gen Identity Security — Independent analyst review

Industry Analysis

- CrowdStrike 2025 Threat Hunting Report: Identity Hunting — Deep dive into identity attack detection techniques
- Cloud Breaches and Identity Hacks Explode — SiliconANGLE analysis of H1 2025 cloud/identity trends
- CrowdStrike Field CTO on Identity-Based Attack Shift — Expert perspective on the identity-first security imperative

Standards & Frameworks

- FIDO2 / WebAuthn Specification — The standard behind phishing-resistant authentication
- NIST SP 800-63B: Digital Identity Guidelines (Authentication) — US government authentication assurance levels
- MITRE ATT&CK: Credential Access Tactics — Comprehensive catalog of credential theft techniques

Related Guides in This Series

- Agent Authentication & Identity — How agents prove their identity (Managed Identity, Entra Agent ID, Foundry Blueprints, MCP auth)
- Agent Protocols: MCP, A2A, ACP — Protocol-level details of how agents connect to tools and each other